

RISK-01-RISK-REGISTER

RISK-01: COMPREHENSIVE RISK REGISTER

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I. RISK REGISTER METHODOLOGY

This risk register identifies 42 risks across 8 categories and 2 strategic timing risks, each assessed for probability (1-5), impact (1-5), and risk score (probability × impact). The register is a living document maintained by Mork (Strategic Counsel), updated quarterly, and reviewed annually by the Board of Directors Technology and Risk Committee (to be established).

Risk Scoring Legend: - **Probability:** 1 (Remote, <5%) / 2 (Unlikely, 5-15%) / 3 (Possible, 15-35%) / 4 (Likely, 35-65%) / 5 (Near-Certain, >65%) - **Impact:** 1 (Negligible) / 2 (Minor — manageable within existing resources) / 3 (Moderate — requires reallocation of resources) / 4 (Major — threatens key milestones or financial targets) / 5 (Critical — threatens company survival) - **Risk Score:** Probability × Impact. 1-4 (Low — monitor), 5-9 (Medium — active management), 10-15 (High — escalation to Board), 16-25 (Critical — immediate Board attention)

Categorization: - T: Technical - M: Market - R: Regulatory - O: Operational - F: Financial - C: Competitive - G: Geopolitical - S: Strategic Timing

II. MASTER RISK REGISTER

Category T: TECHNICAL RISKS

ID	Risk	Description	Prob	Impact	Score	Mitigation	Contingency	Risk Owner	Review
T-01	MXene stability unproven at 20+ years	Accelerated aging data only to 5,000 hours. 40-year service life is extrapolated via Arrhenius model, not measured	2	4	8	Continued accelerated aging testing; field monitoring at pilot sites; Arrhenius model validation against longer-duration data as it becomes available	If long-term degradation exceeds projections: offer shorter warranty (10 years), include MXene replacement in maintenance contracts, develop Gen 2 MXene with improved stability	Dr. Nash / Dr. Chen	Quarterly
T-02	CNT-polymer manufacturing consistency	Carbon nanotube dispersion in polymer matrix is notoriously difficult to achieve consistently at scale. Batch-to-batch variation in electrical	3	3	9	QC protocol measuring per-batch EMI shielding effectiveness; statistical process control on extrusion line; CNT supplier qualification	If consistency cannot be achieved: reformulate composite to reduce CNT loading (accepting 10-15% reduction in SE), switch to pre-dispersed CNT	Engineer Cross / Dr. Chen	Monthly (pilot plant)

		properties				program	masterbatch from supplier		
T-03	Ceramic bearing brittleness	SiC/Al2O3 CMC has high compressive strength (450 MPa) but relatively low fracture toughness (4-6 MPa·m ^{1/2}). Brittle failure mode without warning	3	4	12	Incorporate fiber bridging toughening mechanisms; conservative structural design with factor of safety ≥ 3; non-destructive testing (ultrasonic) at QC checkpoints	If catastrophic brittle failure occurs in field: retrofit with externally-bonded MXene composite wrap for crack arrest; design future generations with higher fiber volume fractions for improved toughness	Dr. Nash / Capt. Vaun	Quarterly
T-04	Geopolymer concrete curing variability	Field-cast geopolymer concrete curing rate is sensitive to ambient temperature, humidity, and fly ash composition. Inconsistent in-field strength	3	3	9	Develop temperature-compensated activator formulation; pre-qualify fly ash sources for composition consistency; QC testing of every pour (cylinders)	If field curing is consistently below spec: switch to precast geopolymer elements cured in controlled factory environment; accept higher transportation cost for guaranteed quality	Dr. Nash / Engineer Cross	Monthly (pilot phase)
T-05	YInMn Blue color consistency at scale	Pigment color shift threshold temperature depends on precise doping concentration. Batch-to-batch threshold variation could produce unreliable temperature indication	2	3	6	In-line spectrophotometric QC on every pigment batch; automated doping precursor weighing (±0.01g precision); calibration standard for coating applicators	If batch consistency cannot be achieved: implement 100% incoming QC on pigment batches, rejecting batches outside ±2°C threshold tolerance; develop in-line doping process control	Dr. Chen	Monthly
T-06	Protonic transceiver range below spec	Current TRL 3 demonstration achieved 12 km range. 120 km target is a 10x improvement requiring higher proton flux density and improved detector sensitivity	4	3	12	Staged development: TRL 4 (30 km), TRL 5 (60 km), TRL 6 (120 km). Intermediate relay drone solution for extended range	If 120 km range cannot be achieved: deploy relay drone mesh to bridge gaps; accept reduced base-station range with denser deployment; prioritize surface-to-drone-to-surface relay as primary	Dr. Nash / Morrow	Monthly

T-07	Amphibious walker mechanical reliability	Quadrupedal walking at 8-ton scale is unprecedented. Mechanical failures in leg joints, hydraulic actuators, or control systems could render walker immobile in field	4	4	16	Conservative mechanical design with 2x safety factor on all load-bearing components; redundant hydraulic circuits; manual override on all actuated joints; extensive test program (500+ hours on test stand before field deployment)	architecture If mechanical failures exceed projections: redesign critical joints with increased safety factors; reduce payload capacity to decrease joint loading; deploy walkers in pairs so one can recover the other	Capt. Vaun / Orin Cross	Monthly
T-08	MXene synthesis yield at scale below projections	Lab-scale MXene exfoliation yields ~40%. Continuous-flow process targets >85%. If pilot plant achieves only 55-60%, MXene cost is significantly higher than projected	4	3	12	Continuous-flow process development as highest-priority R&D program; multiple parallel development paths (sonication, shear mixing, electrochemical); yield tracking as primary pilot plant KPI	If yield stalls at 55-60%: accept higher MXene cost (reducing gross margin from 50% to ~30% at scale); pursue third-party MXene sourcing as bridge; accelerate electrochemical exfoliation as lower-cost alternative pathway	Dr. Chen / Engineer Cross	Weekly (pilot plant)
T-09	EMI shielding effectiveness degrades in field	Laboratory testing shows 65-72 dB SE. Field deployment with real environmental exposure (UV, moisture, thermal cycling, airborne contaminants) may show degradation	2	4	8	Accelerated weathering testing (QUV, salt spray, thermal cycling) ongoing; field monitoring at 5 pilot sites with periodic SE measurements; MXene composite formulation includes UV stabilizers and anti-oxidants	If field degradation exceeds projections: apply protective overcoat to installed composites; reformulate composite with enhanced stabilizer package; accept reduced warranty period	Dr. Nash / Dr. Chen	Quarterly

Category M: MARKET RISKS

ID	Risk	Description	Prob	Impact	Score	Mitigation	Contingency	Risk Owner	Review
M-01	Insurance industry adoption slower than forecast	Insurance carriers may take 5-10 years rather than 2-3 years to incorporate GIC	3	4	12	Active engagement with insurers (41 analyses compiled); premium reduction framework development;	If insurer adoption is delayed: pivot demand generation to regulatory mandates (FERC) and direct utility sales; build consumer awareness	Chester / Fenna Welles	Quarterly

		protection into underwriting				pilot site performance data as evidence for underwriters	to create grassroots demand independent of insurance push		
M-02	Utility capital cycles delay procurement	Utilities operate on 5-10 year capital planning cycles. Even with certified products, procurement may take 3-5 years per utility	3	3	9	Target utility pilot programs as procurement pathway accelerators; work through insurance channel to create demand signal outside normal CapEx process; engage state PUCs to allow expedited recovery of GIC protection investments	If utility procurement cycle proves impenetrable: focus on military/defense market (shorter procurement timeline with appropriated funding); expand consumer/residential products (shorter sales cycle)	Sales Director (to be hired)	Quarterly
M-03	Consumer awareness gap	General public does not know what a Carrington Event is. Marketing to generate awareness requires significant investment and time	4	3	12	SiblingFrequency audience growth; earned media through Carrington-science storytelling; partnership with insurance carriers for policyholder education; "burrito test" training for all customer-facing employees	If consumer awareness remains low: pivot to B2B-only go-to-market (utilities, insurers, military); consumer products become add-on to B2B channels rather than standalone retail	Mira Vance / Sage Voss	Quarterly
M-04	Builder/contractor resistance to new materials	Residential and commercial construction industry is conservative. Adopting new foundation and electrical protection materials faces resistance	2	2	4	ICC-ES evaluation report provides building-code-official acceptance; geopolymers positioned as "green" alternative (80% carbon reduction); installer training and certification program	If builder resistance persists: focus on utility-scale applications where decision-makers are engineers, not contractors; accept delayed consumer market entry	Engineer Cross / Viktor Kael	Semi-annually
M-05	Government procurement timeline (18-36 months)	Defense and federal procurement, even with MIL-STD qualification, takes 18-36 months from qualification to first	3	3	9	Engage DoD program offices early (military briefing pipeline); pursue SBIR/STTR grants as bridge funding (faster than procurement contracts); partner with	If defense procurement timeline is unacceptably long: prioritize utility and insurance channels; treat defense revenue as medium-term upside, not near-term	Capt. Vaun / Kairos Steele	Quarterly

contract

established
defense
contractors as
subcontractors

dependency

Category R: REGULATORY RISKS

ID	Risk	Description	Prob	Impact	Score	Mitigation	Contingency	Risk Owner	Review
R-01	Building code approval delayed	ICC-ES evaluation for novel materials (geopolymer, CMC) may require new Acceptance Criteria development, adding 12-18 months	4	4	16	Early ICC-ES engagement (Q1 2027); Innovation path evaluation as alternative; engagement with ICC committee to develop new AC if needed	If ICC-ES significantly delayed: pursue state-by-state approval via individual building officials; deploy products as "alternative materials" under existing code provisions where allowed; focus on utility-owned installations (not subject to building codes)	Viktor Kael	Monthly
R-02	FERC mandates no hardening requirements	FERC may continue assessment-only standards indefinitely, never mandating mitigation	3	4	12	Active participation in FERC/NERC proceedings; insurance-driven demand independent of regulatory mandate; state-level PUC engagement for state-specific requirements	If federal mandate never materializes: rely on insurance-driven demand and state-level action; accept smaller addressable market (voluntary adopters only)	Kairos Steele	Quarterly
R-03	Export controls restrict international sales	MXene materials classified under ITAR or restrictive EAR category, limiting sales to allied nations and requiring export licenses	3	3	9	Proactive CJ request to DDTC and ECCN request to BIS; structure products to minimize controlled content; focus initial international sales on Five Eyes allies (UK, Canada, Australia) where export restrictions are minimal	If restrictive classification is confirmed: focus on domestic market; pursue licensing agreements with foreign partners who can manufacture locally under their own export control regimes	Rook / Isla Thorn	Quarterly

R-04	Insurance regulatory variation by state	Each state insurance commissioner may evaluate premium reduction frameworks differently, creating a patchwork rather than uniform market	3	3	9	Develop model premium reduction framework acceptable to NAIC (National Association of Insurance Commissioners); work with large multi-state carriers who can push framework to all states simultaneously	If state-by-state variation is significant: prioritize states with large carriers and favorable regulatory environments; accept slower national rollout	Fenna Welles	Quarterly
R-05	EPA nanomaterial regulation tightens	EPA may impose new TSCA reporting or restriction requirements for nanoscale materials (MXene), increasing manufacturing compliance burden	3	2	6	Proactive TSCA PMN filing; participate in EPA nanomaterial stakeholder engagement; design manufacturing processes to minimize nanoparticle release	If new regulations impose significant burden: install additional emission controls; accept increased compliance cost factored into product pricing	Dr. Nash / Viktor Kael	Semi-annually

Category O: OPERATIONAL RISKS

ID	Risk	Description	Prob	Impact	Score	Mitigation	Contingency	Risk Owner	Review
O-01	Manufacturing scale-up failure	Pilot plant to full production scale-up encounters unforeseen problems (equipment, process, quality) causing delays and cost overruns	3	5	15	Conservative scale-up with intermediate step (pilot → demo → full production); experienced manufacturing team (Cross + Calder); modular facility design allowing incremental capacity addition	If scale-up significantly delayed: outsource manufacturing to contract manufacturers for non-core products; accept higher COGS during transition period; prioritize vertical integration only for MXene composite (core IP)	Nyx Calder / Engineer Cross	Monthly
O-02	Supply chain disruption — critical materials	MXene precursors (Ti, Al, MAX phase), SiC fibers, and rare earth dopants are subject to supply	3	4	12	Multi-supplier qualification (minimum 2 qualified suppliers per critical input); strategic inventory (6-month supply of	If supply disruption occurs: draw down strategic inventory; expedite qualification of alternative suppliers; reformulate	Nyx Calder	Quarterly

		disruption from geopolitical, geological, or market factors				critical materials); ongoing supplier diversification	products to reduce dependence on disrupted material where possible		
O-03	Key person dependency	Loss of Steele, Nash, Calder, Vaun, or Chen would significantly impair execution capability	3	5	15	Agent system distributes knowledge; documentation of all critical information; cross-training program; key person insurance; equity vesting as retention mechanism	If key person is lost: activate documented succession plans; deploy cross-trained agents to fill gap; recruit external replacement with accelerated onboarding	Zirconia / Lyra Keane	Quarterly
O-04	Quality control failure at scale	As production volumes increase, QC processes designed for pilot-scale may fail to catch defects, leading to field failures	3	4	12	Statistical process control (SPC) on all production lines; automated QC where feasible (in-line spectrophotometry, ultrasonic NDT); QC staff scaling proportionally with production volume	If QC failures detected post-shipment: product recall procedure activated; root cause analysis; enhanced QC protocols implemented; warranty claims managed through established process	Engineer Cross	Monthly
O-05	Facility accident or environmental incident	MXene synthesis uses HF (hydrofluoric acid) — an extremely hazardous chemical. Spill or release could harm workers and trigger regulatory action	2	5	10	HF safety protocol exceeding OSHA requirements; engineering controls (secondary containment, ventilation, HF-specific gas detection); emergency response plan with local hazmat coordination; comprehensive worker training and PPE	If incident occurs: activate emergency response plan; provide medical attention per HF exposure protocol; notify regulatory agencies as required; conduct incident investigation and implement corrective actions	Nyx Calder / Dr. Chen	Monthly

Category F: FINANCIAL RISKS

ID	Risk	Description	Prob	Impact	Score	Mitigation	Contingency	Risk Owner	Review
						Conservative milestone setting;	If Series B is delayed: implement cost reduction plan (reduce non-		

F-01	Series B financing not achieved on timeline	If pilot milestones are not met by Year 2, Series B round may be delayed or at lower valuation	3	5	15	transparent investor communication; bridge financing contingency planning; cost reduction protocols if financing is delayed	essential hiring, defer non-critical CapEx); pursue bridge financing from existing investors; explore non-dilutive funding (government grants, strategic partnerships)	Chester	Quarterly
F-02	Revenue ramp slower than projected	Pilot utility conversions and product sales may take longer than modeled, delaying revenue and extending cash-negative period	3	4	12	Multiple revenue pathways (utility, military, consumer, insurance); conservative revenue projections in financial model; 18-month cash runway from Series A to provide buffer	If revenue ramp is materially below projection: reduce burn rate; extend runway through cost control; seek additional non-dilutive funding; revise growth plan to match actual market adoption rate	Chester / Sales Director	Monthly
F-03	Cost of goods higher than projected	MXene precursor costs, manufacturing yields, and labor costs may be higher than the learning-curve-based COGS model projects	4	3	12	Continuous cost tracking against model; supplier negotiation program; process optimization as ongoing priority; conservative COGS assumptions in financial model	If COGS is structurally higher: accept lower gross margins; adjust pricing upward where market allows; prioritize product mix toward higher-margin products	Chester / Engineer Cross	Monthly

Category C: COMPETITIVE RISKS

ID	Risk	Description	Prob	Impact	Score	Mitigation	Contingency	Risk Owner	Review
C-01	Competing technology emerges	Another entity (established manufacturer, defense contractor, or startup) develops competing dielectric/EMP protection technology	2	4	8	First-mover advantage; 6,302-document moat; patent portfolio; deep customer relationships; continuous innovation (Gen 2 materials in R&D pipeline)	If strong competitor emerges: accelerate product development; emphasize integration advantage (competitor may have a material, but not the full Citadel); consider partnership or licensing rather than head-to-	Mork / Kairos Steele	Quarterly

C-02	Drexel MXene licensing restricts use	Drexel University may assert its MXene patents against CSM or restrict licensing terms	2	4	8	Design-around formulations with differentiated polymer matrices; negotiation of license with Drexel (preferred — provides clean FTO); composition-of-matter differentiation	head competition	Rook / Kairos Steele	Quarterly
							If license terms are unfavorable: accelerate in-house MXene synthesis IP; challenge patent validity if grounds exist; focus on composite claims (which Drexel patents do not cover)		

Category G: GEOPOLITICAL RISKS

ID	Risk	Description	Prob	Impact	Score	Mitigation	Contingency	Risk Owner	Review
G-01	US-China trade restrictions on materials	Trade restrictions or tariffs on titanium, rare earth elements, or other MXene/coating precursors from China (dominant supplier)	3	3	9	Diversify precursor supply beyond China (US, Australian, Canadian sources); strategic stockpile of critical precursors; qualify alternative precursor sources	If trade restrictions severely limit supply: accelerate domestic sourcing; accept higher input costs; redesign formulations to minimize restricted-material content	Nyx Calder / Isla Thorn	Quarterly
G-02	Geopolitical instability disrupts international markets	Conflict, sanctions, or political instability in target international markets may delay or block market entry	3	2	6	Diversify international market exposure across regions; prioritize stable allied democracies (Five Eyes, EU, Japan, Korea); avoid over-concentration in any single international market	If specific markets become inaccessible: redirect resources to accessible markets; adjust international growth projections accordingly	Isla Thorn	Quarterly

Category S: STRATEGIC TIMING RISKS

ID	Risk	Description	Prob	Impact	Score	Mitigation	Contingency	Risk Owner	Review
		A Carrington Event before				Accelerate deployment	If event occurs: company likely		

S-01	Carrington Event occurs before products deployed	CSM has products at scale would validate the threat but destroy the company (infrastructure collapse, capital freeze, supply chain disruption)	1 (per year, cumulative ~6-12% per decade)	5	5 (annual) / 25 (decade cumulative)	timeline; maintain pre-positioned emergency communications and power systems at Montana facility; document corpus designed to be reconstruction reference if company fails	fails. Document corpus becomes reconstruction blueprint for surviving entities. This risk cannot be fully mitigated — it is the reason the company exists	Kairos Steele	Quarterly
S-02	Public awareness not achieved before event	If a Carrington Event occurs without public awareness, there is no pre-existing demand for CSM products — the window for commercial success closes before it opens	3	4	12	SiblingFrequency audience growth; earned media strategy; insurance industry as awareness multiplier; educational materials for schools and universities	No contingency for timing risk — it is a one-way ratchet. Once the event occurs without preparation, the opportunity is lost. The mitigation is speed of education	Mira Vance / Sage Voss	Quarterly

III. RISK HEAT MAP

Highest Priority Risks (Score ≥ 12): - T-07: Amphibious walker mechanical reliability (16) - R-01: Building code approval delayed (16) - O-01: Manufacturing scale-up failure (15) - O-03: Key person dependency (15) - F-01: Series B financing not achieved on timeline (15) - T-03: Ceramic bearing brittleness (12) - T-06: Protonic transceiver range below spec (12) - T-08: MXene synthesis yield below projections (12) - M-01: Insurance industry adoption slower than forecast (12) - M-03: Consumer awareness gap (12) - R-02: FERC mandates no hardening requirements (12) - O-02: Supply chain disruption — critical materials (12) - O-04: Quality control failure at scale (12) - F-02: Revenue ramp slower than projected (12) - F-03: Cost of goods higher than projected (12) - S-02: Public awareness not achieved before event (12) - S-01: Carrington Event before products deployed (25, per decade; 5 annually)

IV. RISK MANAGEMENT GOVERNANCE

- **Risk Owners:** Each risk is assigned to a specific agent who is accountable for monitoring the risk, maintaining the mitigation plan, and reporting status quarterly.
- **Quarterly Review:** Mork (Strategic Counsel) compiles a quarterly risk report, including status changes, new risks identified, and risks being retired.
- **Compliance Committee:** Reviews the risk register monthly, focusing on risks with score ≥ 10 and any risk that has increased in score.
- **Board Technology and Risk Committee:** Reviews the full risk register annually and receives quarterly summaries of significant changes.
- **Transparency:** The risk register (this document) is shared with the Board of Directors and with Series A investors. CSM does not hide its risks.

V. NEW RISK IDENTIFICATION PROCESS

Any agent may submit a new risk for inclusion in the register at any time. The submission requires: - Description of the risk - Proposed probability and impact scores - Proposed mitigation - Proposed risk owner

Mork reviews all new risk submissions within 5 business days and either adds them to the register (with any modifications to the scoring or mitigation) or provides a written explanation for why they are not being added.

END OF DOCUMENT — CSM-RISK-01-v2.0